

28th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2024)

Privacy-Oriented Feature Selection for Multivariate Time Series Classification

Adrian-Silviu Roman^{a,*}, Béla Genge^a, Roland Bolboacă^a

^a “George Emil Palade” University of Medicine, Pharmacy, Science and Technology of Targu Mures, Targu Mures, Mures, 540142, Romania

Abstract

The widespread use of sensors in various domains such as automotive or healthcare has greatly increased the amount of collected data, leading to important privacy issues. Although the sensor data does not directly identify individuals, it can still reveal sensitive information. Addressing privacy in the context of Time Series Classification (TSC) presents challenges, including the need to balance data utility and privacy protection. The current study introduces a novel privacy-oriented feature selection methodology for TSC, aiming to improve data privacy while preserving utility. We propose a dual-model approach, leveraging two classifiers with opposing objectives: a Utility-Focused Classifier (UFC) and a Privacy-Breaking Classifier (PBC). The methodology introduces the Importance Difference Score (IDS) for feature ranking with the objective of selecting features important for the UFC while removing the features essential for the PBC. The approach includes two feature clustering techniques, one based on IDS and the other on K-means clustering, to optimize the feature selection process. The experiments performed on two driving datasets and two Human Activity Recognition (HAR) datasets, evaluate the effectiveness in reducing the accuracy of potential adversarial classifiers while maintaining appropriate levels of utility. We contribute to the state-of-the-art by offering a configurable framework for feature selection to balance the privacy and utility of the data in TSC.

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Peer-review under responsibility of the scientific committee of the 28th International Conference on Knowledge Based and Intelligent information and Engineering Systems

Keywords:

utility-preserving data privacy; feature selection; feature clustering; time series classification

1. Introduction

Data collected from sensor-based systems and devices, which are omnipresent in our daily lives, provides vital insights for various fields, such as transportation, healthcare, or smart city management. These sensors, ranging from simple temperature monitors to complex accelerometers in smartphones, continuously gather data about user activities and behaviors. Such data is typically transmitted to centralized databases, enabling analysts to extract and analyze

* Corresponding author.

E-mail address: adrian.roman@umfst.ro

meaningful information. However, this process raises significant privacy concerns. As sensor data, often processed as time series data, is sent to third-party processors for analysis, there is a potential risk that sensitive and private information is exposed. Although sensor data might not directly contain personal identifiers like names or emails, they may disclose user-related sensitive information (e.g., behaviors or health metrics) and lead to user identification or re-identification [6, 17]. Thus, protecting sensor data from both “honest-but-curious” data analysts and attackers is necessary.

Protecting the privacy of multivariate time series data poses significant challenges due to aspects such as feature correlation and a large number of features. Furthermore, time series data collection can vary, ranging from finite datasets to infinite data streams. To address privacy concerns associated with publishing time series data, various techniques have been proposed, with a significant focus on approaches involving [Differential Privacy \(DP\)](#) [5, 11, 14]. These methods apply perturbation methods to the time series, making it difficult to accurately reconstruct specific values solely based on timestamps.

Despite the effectiveness of these approaches in preserving privacy, the introduction of noise to achieve this can significantly distort the original values, thus reducing data utility. In the field of [Time-Series Classification \(TSC\)](#), data privacy predominantly relies on two primary methodologies. Firstly, securing input data through various means such as perturbation, encryption, de-identification, data transformation, and the utilization of machine learning techniques like deep autoencoders [13, 19]. Secondly, the use of classification models designed to maintain privacy, thereby hiding training data against potential adversaries [1, 2]. However, a notable portion of existing methodologies [1, 13] overlooks the presence of an adversarial model characterized by a powerful classifier.

Implementing perturbation or other forms of privacy protection does not provide foolproof privacy guarantees. While [DP](#) demonstrates effectiveness in aggregate data publishing scenarios, its applicability to time series data release is limited. In the context of [TSC](#), these limitations are more prominent, as powerful classifiers are capable of potentially identifying or reidentifying a user despite privacy measures. Furthermore, in certain areas such as the medical field, releasing time series data in a modified form may not be viable, as data precision has critical importance.

The presented methodology proposes feature selection as a necessary step for achieving data privacy when releasing multivariate time series data. We introduce a privacy-oriented feature selection approach that not only satisfies privacy objectives, even in the context of [TSC](#), but also preserves data utility. This mechanism is designed to be configurable and to allow data owners to achieve the desired balance between privacy and utility. In summary, the research presents the following important contributions to the state-of-the-art.

- It defines the problem of achieving privacy in the context of [TSC](#), using a dual-model, consisting of two competing classifiers, an [Utility-Focused Classifier \(UFC\)](#) and a [Privacy-Breaking Classifier \(PBC\)](#);
- It introduces the [Importance Difference Score \(IDS\)](#), designed for feature ranking, to measure the significance of a feature for [UFC](#) compared to [PBC](#);
- It presents a methodology for selecting the features that are more important for [UFC](#) than for [PBC](#), with the objective of decreasing the classification accuracy of the [PBC](#) while maintaining the accuracy of [UFC](#) to reasonable levels. The method relies on two features clustering techniques based on importance coefficients, the first one utilizing [IDS](#), while the second one, based on the K-means clustering;
- It evaluates the effectiveness of the proposed methodology on two drivesets [20, 12] and two [Human Activity Recognition \(HAR\)](#) datasets [16, 18].

The remainder of this paper is structured as follows. Section 2 provides an overview of the state-of-the-art focused on feature selection. Further, Section 3 describes the addressed problem. Section 4 presents an in-depth description of the proposed approach. This is followed by experimental results in Section 5. The paper concludes in Section 6.

2. Related Work

In the multivariate time series classification domain, feature selection entails the selection of relevant features to improve algorithm accuracy, reduce computational complexity, and enhance interpretability [4]. Beyond its primary objectives, feature selection has also been explored as a privacy-enhancing tool.

Zhang et al. [23] proposed a feature selection technique utilizing feature importance in a differentially private manner. It aims to reduce data correlation and correlated sensitivity, thereby improving the utility of the data. The approach relies on a threshold selection mechanism, which is critical for determining the importance of features while ensuring differential privacy. The utility of feature selection for privacy protection was further demonstrated by Jafer et al. [8]. Their research demonstrated that even without explicit privacy considerations, the process of eliminating irrelevant or redundant attributes could improve privacy. However, this method lacks explicit privacy-aware considerations during the feature selection process. Addressing the need for a privacy-aware feature selection, Jafer et al. [9] introduced an automatic feature selection framework that considers both privacy and utility objectives simultaneously and utilizes a two-dimensional measure. The adversarial model considered in their study is based on the *uninformative principle*, suggesting that attackers gain minimal information about sensitive data from the released dataset. In the context of collaborative data classification, Sheikhalishahi and Martinelli [21] proposed a framework that highlights privacy protection for data providers willing to share their data for collaborative benefits. This framework introduces a privacy mechanism that eliminates irrelevant features, thereby balancing accuracy and privacy to maintain data utility for classification. Sriyayanthi and Sethukarasi [22] extended the concept of privacy-preserving in collaborative data sharing by proposing a model that incorporates clustering, anonymization, and feature selection techniques. The model begins with the identification of relevant features using *Symmetrical Uncertainty (SU)* to pinpoint attributes contributing significantly to data utility. It then applies Spearman's Correlation Coefficient to eliminate redundant features, effectively reducing data dimensionality and enhancing privacy without compromising data utility for classification purposes.

Previous works demonstrated privacy-oriented methodologies for feature selection, ranging from differential privacy approaches to automatic selection frameworks. Our research introduces a dual-model approach that directly takes into consideration adversarial threats capable of compromising user privacy. Unlike previous studies that either focus on enhancing utility or applying privacy measures without considering specific adversarial models, our method employs two opposing classifiers: one optimizes for utility by selecting essential features for classification accuracy, and the other is utilized for removing features that could be utilized by potential attackers. The approach places both privacy and utility as core elements of the feature selection process.

3. Problem Definition

Consider the data collected from sensor-based devices in the form of multivariate time series, transmitted to a central database, and then shared with third-party processors for classification. Within this scenario, two types of classifiers are examined: (i) a **UFC**, aimed at identifying patterns within the collected data, and (ii) a **PBC**, designed to expose the identities of users within the data. The objective of our research is to propose a method for feature selection such that the classification accuracy of the **PBC** is reduced, for increased privacy. At the same time, the classification accuracy of the **UFC** or **UFCs** should remain high enough to maintain the overall utility.

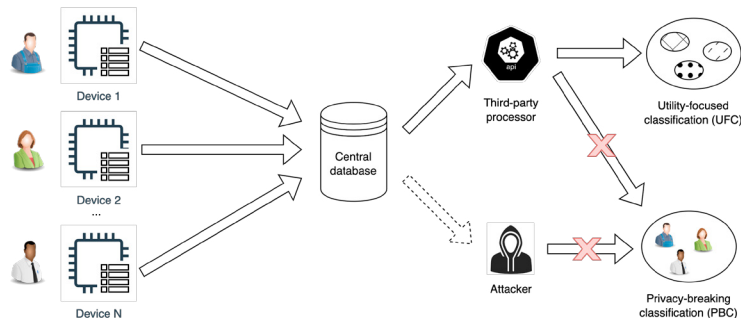


Fig. 1: Problem description. Multivariate time series data, collected from several sensor-based devices, is transmitted to a centralized database. The selected features should decrease the accuracy of **PBC**, while maintaining the accuracy of the **UFC**.

Example 1. Consider a fleet management company that collects data from sensors in their vehicles to optimize fleet operations and improve driver safety. The data includes information such as vehicle speed, engine diagnostics, and fuel consumption. This data is transmitted to the company's servers for analysis and decision-making purposes. The

company has two primary objectives: (i) to classify driving behavior, distinguishing between normal or aggressive driving, and (ii) to monitor vehicle health and identify potential maintenance issues. These objectives do not involve personal information or identification. However, the company must prevent potential threats such as unauthorized access to vehicle data or the development of **PBC** models capable of identifying drivers. The goal is to utilize the collected data to optimize fleet management while minimizing the risk of privacy breaches.

Example 2. Similarly, consider wearable fitness trackers that collect data on human activity and transmit it to a health monitoring service. The data is gathered by sensors embedded in the wearable device and sent to the service provider's servers via a mobile app. The health monitoring service aims to accurately classify different types of physical activities performed by the user, such as walking, running, or cycling. However, the service provider must protect the user's privacy against potential threats, such as unauthorized access to the collected data or the development of **PBC** models capable of identifying individuals based on their activity patterns. Thus, the goal is to ensure that the collected data can be effectively utilized for activity recognition while minimizing the risk of user identification.

The process of training the **UFC** model assumes a secure environment. In both the training and testing phases of constructing the **UFC**, data labels for each batch of records are stored in the data warehouse. Afterward, there is the possibility to remove the data labels from the database for privacy purposes. No label information is shared with third-party systems. We consider the following privacy-breaching scenarios (the adversarial model): (i) the “honest-but-curious” entities, such as a data analyst who has access to all the data needed to construct a classifier that compromises privacy; (ii) an attacker that is capable of constructing a **PBC** for identifying the users by getting access to various data sources (training dataset with labels, the unlabeled data collected from the devices, and other observable data).

4. Proposed Approach

Consider the continually collected data, consisting of records with n features, transmitted to a central location for processing. Let $\mathbf{X}_i^t$ be the time series data collected on the sensor-based device i , at time t . $\mathbf{X}_i^t$ is represented as a vector of n elements, each element denoting measurements associated with a particular sensor.

TSC involves assigning a label or category to a given time series data based on its inherent patterns, trends, or features. A classifier model, denoted as $f : \mathbb{R}^n \rightarrow C$, where $C = \{1, \dots, c\}$, is applied to any input $\mathbf{X}_i^t$. The function $f(\mathbf{X}_i^t)$ provides an estimation of the category to which $\mathbf{X}_i^t$ belongs [7]. Let f_p and f_u represent two classification functions, where $f_p : \mathbb{R}^n \rightarrow C_p$ and $f_u : \mathbb{R}^n \rightarrow C_u$ output the class to which $\mathbf{X}_i^t$ belongs. In the case of **PBC**, $f_p(\mathbf{X}_i^t)$ identifies the set of collected values as belonging to a user from a specified class C_p , compromising user privacy. Conversely, for the **UFC**, the function $f_u(\mathbf{X}_i^t)$, categorizes non-sensitive information from a class C_u .

Let $\mathbf{Y}_{pi}^t$ denote the label information for user identification, while $\mathbf{Y}_{ui}^t$ represents the label for classifying non-user-related data. The classification accuracy, which shows the number of correct classifications, for the two classifiers f_u and f_p is defined as follows:

$$\mathcal{A}_u = \frac{1}{N \cdot T} \sum_{i=1}^N \sum_{t=1}^T \mathbb{I}(\mathbf{Y}_{ui}^t = f_u(\mathbf{X}_i^t)), \mathcal{A}_p = \frac{1}{N \cdot T} \sum_{i=1}^N \sum_{t=1}^T \mathbb{I}(\mathbf{Y}_{pi}^t = f_p(\mathbf{X}_i^t)), \quad (1)$$

where $\mathcal{A}_u$ is the accuracy of f_u , $\mathcal{A}_p$ is the accuracy of f_p , T the number of data records collected, N is the number of sensor-based devices, and $\mathbb{I}(c)$ the binary indicator function returning 1 iff the condition c is true, and 0 otherwise.

Further, consider the collection of available features of the dataset, denoted as $\mathcal{F} = \{F_1, \dots, F_j, \dots, F_n\}$. The objective of proposed methodology is to identify a subset $\mathcal{F}^*$ of $\mathcal{F}$ with n^* features, $n^* < n$, such that:

$$\mathcal{A}_p^* \ll \mathcal{A}_p \text{ and } \mathcal{A}_u^* \approx \mathcal{A}_u, \quad (2)$$

where $\mathcal{A}_p^*$ and $\mathcal{A}_u^*$ are the classification accuracy values calculated similarly to Eq. 1, for the selected features $\mathcal{F}^*$.

The proposed method for identifying the possible feature subsets $\mathcal{F}^*$ consists of the following steps: (i) compute feature importance for the two classifications (UFC and PBC) using all available features in $\mathcal{F}$; (ii) cluster features based on the computed importance coefficients; (iii) select the features in $\mathcal{F}^*$, features there are important for UFC while excluding the features that determine a high classification accuracy for PBC; (iv) validate the selected features, taking into account the desired privacy and utility objectives (i.e., values of $\mathcal{A}_p^*$ and $\mathcal{A}_u^*$).

4.1. Feature Importance Computation

The computation of feature importance [3] aims to evaluate the relative significance of features within a dataset for a particular classification task. An advantage of utilizing feature importance computation techniques is their ability to assign a feature score without requiring prior knowledge of the adversarial model. Thus, this computation relies solely on available data labels and anticipates potential classifications.

Let $\mathbf{X}$ be the set of all $\mathbf{X}_i^t$ selected for the training phase with all available features in $\mathcal{F}$. By computing the feature importance for classifying $\mathbf{X}$ on class C_p using the labels $\mathbf{Y}_p$ and for classifying $\mathbf{X}$ on class C_u using labels $\mathbf{Y}_u$, we obtain I_u and I_p , the vectors of importance scores, $I_u = [i_1^u, i_2^u, \dots, i_d^u]$ and $I_p = [i_1^p, i_2^p, \dots, i_d^p]$, with $i_j^u, i_j^p \in [0, 1], \forall j \in \{1, \dots, d\}$.

We consider the scenario when no prior knowledge exists on the classification model used for UFC or PBC. For brevity, we only propose using the Random Forest (RF) algorithm [3], a popular technique for feature ranking and feature importance computation, to identify the features that hold the most relevance for classifications. For the current research, we selected the impurity-based feature importance (e.g., Gini importance [3]). Other scenarios may be considered for future research when targeted UFC and PBC are known. In this case, techniques such as permutation feature importance or Shapley Additive Explanations (SHAP) may be utilized. The only restriction our methodology imposes is to quantify the relative importance of each feature.

4.2. Importance-based Feature Clustering

The next step in identifying $\mathcal{F}^* \subset \mathcal{F}$ is to cluster $\mathcal{F}$ into two distinct subsets based on the calculated importance scores, I_p and I_u . These subsets are: $\mathcal{F}_u$, including features essential for UFC tasks, and $\mathcal{F}_p$, with features that play a significant role in resolving the PBC, such that $\mathcal{F} = \mathcal{F}_u \cup \mathcal{F}_p$. In the case of one UFC and one PBC, considered here, $\mathcal{F}^* = \mathcal{F}_u$.

The first clustering approach proposed involves the computation of the Importance Difference Score (IDS), defined as $S_{F_j}(i_j^u, i_j^p) = i_j^u - i_j^p$, $i_j^u \in I_u$ and $i_j^p \in I_p$, for each feature F_j . I_u and I_p are the importance coefficients calculated for utility-focused and privacy-breaking classifications, respectively. The following rules for clustering are considered:

$$F_j \in \mathcal{F}_u \text{ if } S_{F_j}(i_j^u, i_j^p) \geq \theta_I, \text{ and, } F_j \in \mathcal{F}_p \text{ if } S_{F_j}(i_j^u, i_j^p) < \theta_I, \quad (3)$$

where θ_I is the decision boundary. The score S_{F_i} determines when a feature shows a significantly greater significance in a classification context compared to another. Fig. 2 exemplifies a group of ten features, their feature importance coefficients for an UFC and a PBC, and possible clustering of the features based on rules stated above, for $\theta_I = 0.0$ and $\theta_I = 0.05$.

The second proposed clustering method involves using the K-means method [15], a widely utilized clustering algorithm. Consider $\mathcal{O}$ a set of n two-dimensional points o_j , where n is the number of features. Each point o_j is a pair of values (i_j^u, i_j^p) from vectors I_u and I_p , representing the importance coefficients of feature F_j . The objective is to determine k disjoint clusters $\{C_1, \dots, C_k\}$, $k \geq 2$, with centroids $\{\mu_1, \dots, \mu_k\}$, such that the total *within-cluster sum of square* (WCSS) distances between each point and the mean of its cluster is minimized.

The features are clustered into two distinct groups: those that are important for UFC and those essential for PBC. We refine the clustering approach by selecting initial centroids to reflect the objective of separating representative features for UFC and PBC. We begin by selecting two initial centroids for the clustering process, specifically from the set $\mathcal{O}$. These centroids are chosen as follows: one centroid is selected for its maximum value in I_u , indicating its high

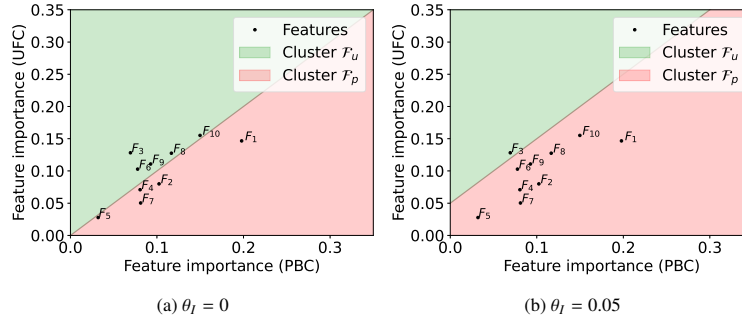


Fig. 2: Examples of feature clustering based on IDS computation for two classifications (UFC and PBC).

Algorithm 1: Feature selection using importance-based K-means clustering for one **UFC** and one **PBC**

Input: $\mathcal{F}$ (vector of available features), I_u (vector of feature importance coefficients for UFC), I_p (vector of feature importance coefficients for PBC), k (number of clusters);

Output: The vector of selected features $\mathcal{F}^*$;

Function `kmeansFeatureSelection`($\mathcal{F}$, I_u , I_p , k):

```

    index_max_iu ← indexMaxValue( $I_u$ ); // get the indexes of
    index_max_ip ← indexMaxValue( $I_p$ ); // the largest values in  $I_u$  and  $I_p$  - Section 4.1
     $\mu^* \leftarrow (I_u[\text{index\_max\_iu}], I_p[\text{index\_max\_iu}]);$  // select two initial clusters - Equation 4
     $\mu^{**} \leftarrow (I_u[\text{index\_max\_ip}], I_p[\text{index\_max\_ip}]);$ 
     $C, \mu \leftarrow kMeans(\mathcal{F}, k, \mu^*, \mu^{**});$  // compute K-means clusters and centroids - Section 4.2
     $\mathcal{F}^* \leftarrow C[\text{indexMaxDistance}(\mu)];$  // select the cluster with the centroid that lies at
    the maximum distance (Equation 5) above the decision boundary
    return  $\mathcal{F}^*$ ;

```

End Function

importance to **UFC**, and the other for its maximum value in I_p , highlighting its importance in **PBC**:

$$\mu^* = \arg \max_{(i_j^u, i_j^p) \in O} i_j^u \text{ and } \mu^{**} = \arg \max_{(i_j^u, i_j^p) \in O} i_j^p. \quad (4)$$

After the completion of the clustering process, the most significant cluster for **UFC** is selected. To identify this cluster, denoted as $\mathcal{F}_u$, the distances from the centroids to the decision boundary defined by condition $i_j^u = i_j^p$ are computed. The centroid of the cluster $\mathcal{F}_u$, denoted $\mu_u = (x_u, y_u)$, with $x_u \geq y_u$, lies at the maximum distance, defined as:

$$d_{\max} = \max \left(\frac{|x_i - y_i|}{\sqrt{2}} \right), \forall \mu_i = (x_i, y_i), i \in \{1, 2, \dots, k\}. \quad (5)$$

This procedure ensures that the chosen cluster $\mathcal{F}_u$, if it exists, includes the features favoring the **UFC**. Algorithm 1 summarizes the steps described above and Fig. 3 exemplifies a group of ten features, their feature importance coefficients for an **UFC** and a **PBC**, and a K-means clustering of the features, for $k = 2$ and $k = 3$.

4.3. The Case of Multiple Utility Objectives

The scenario involving multiple **UFCs** deserves particular consideration. As presented in *Example 1*, the data owner may be interested in multiple utility objectives. In such instances, the proposed approach can be adapted accordingly.

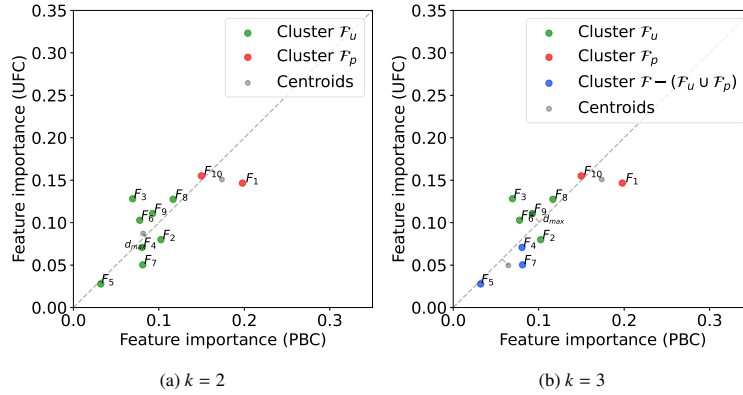


Fig. 3: Examples of K-means-based feature clustering for two classifications (UFC and PBC). d_{max} is computed for selecting $\mathcal{F}_u$.

Let U be the number of targeted **UFCs**. For the **IDS**-based feature clustering approach, consider $\mathcal{F}_{uj}$ the set of selected features for j -th **UFC**. $\mathcal{F}_{uj}$ is the set of features important for the j -th **UFC**. The final set of selected features is the union of all $\mathcal{F}_{uj}$, $\mathcal{F}^* = \bigcup_{j=1}^U \mathcal{F}_{uj}$.

In the case of K-means clustering, at least $U + 1$ clusters will be computed (U clusters for **UFCs** and one for **PBC**) with $U + 1$ initial centroids computed according to Eq.4. After the K-means algorithm completes the clustering, U clusters are selected, one for each **UFC**, by computing the maximum distance from the cluster to the decision boundary between the **UFC** cluster and the **PBC** cluster, according to Eq.5. Finally, the union of features from the U clusters forms $\mathcal{F}^*$. Fig. 4 shows the clusters resulted in the case of two **UFCs**, for $k = 3$ and $k = 4$.

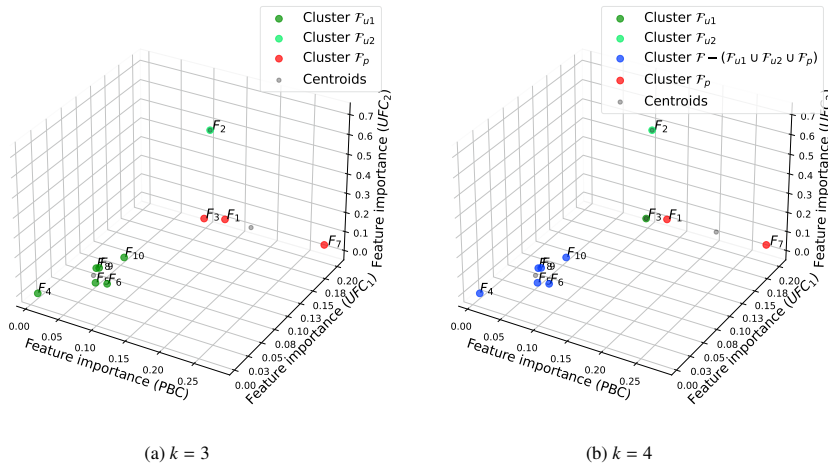


Fig. 4: Examples of K-means-based feature clustering for multiple **UFCs**. In this case, $\mathcal{F}^* = \mathcal{F}_{u1} \cup \mathcal{F}_{u2}$.

5. Experimental Results

We evaluated our proposed methodology using four datasets, divided into two categories: first, two datasets with data collected from autonomous vehicles (UAH-Driveset [20], HCRL Driving Dataset [12]), and second, two datasets focused on **HAR** (MotionSense [16], PAMAP2 [18]). The number of initial features selected for each dataset is listed in Table 1.

5.1. Experiments

In the initial phase of our experiments, we evaluated the classification accuracy using the original set of features $\mathcal{F}$ to address both utility and privacy goals. The experiments were conducted using three distinct classification models: firstly, an LSTM-based technique, called FCN-LSTM (introduced by Karim *et al.* [10]), a state-of-the-art approach in the field of driver identification [6]; secondly, a [Support Vector Machines \(SVM\)](#) classifier, a marginal classifier which maximizes the margin between data samples belonging to different classes; and thirdly, an XGBoost classifier, known for its efficiency and effectiveness in handling various types of data. During the data preprocessing phase, all datasets were normalized using a standard scaler to achieve a mean of 0 and a standard deviation of 1 for every feature. Additionally, we divided the data, allocating 70% for training purposes and the remaining 30% for testing. The results (Table 1) showed considerable accuracy levels across all employed classification techniques, even though the [SVM](#) classifier demonstrated relatively lower performance.

Table 1: Classification results for the initial set of features

Dataset	# of features in $\mathcal{F}$ (initial set of features)	Classification objective	Classification accuracy		
			FCN-LSTM	SVM	XGBoost
UAH Driveset [20]	17	Driver (PBC)	0.9156	0.7435	0.9770
		Behavior (UFC)	0.9462	0.8457	0.9785
		Road (UFC)	0.9999	0.9993	0.9999
HCRL [12]	15	Driver (PBC)	0.9130	0.7733	0.9288
		Road (UFC)	0.9636	0.9281	0.9681
MotionSense [16]	12	User (PBC)	0.8592	0.7066	0.7782
		Activity (UFC)	0.9154	0.8359	0.8366
PAMAP2 [18]	31	User (PBC)	0.9980	0.9997	1.00
		Activity (UFC)	0.9653	0.9678	0.9973

Further, the experiments involved the computation of feature importance for each classification objective, as explained in Section 4.1. Thus the relevance of each feature for user/driver detection and non-user-focused data classification (behavior, road type, activity type) was calculated. For instance, for the UAH dataset, the importance coefficients were computed for the *driver* on the privacy side, and, for *behavior* and *road type*, for the utility side. Vectors containing the importance coefficients for each feature (I_u, I_p) were the output of this processing stage.

Next, features were clustered based on the rules outlined in Section 4.2, taking into account the computed feature importance coefficients determined in the preceding processing stage. Two clustering approaches were evaluated: one utilizing [IDS](#) values with $\theta_l = 0$, and the other employing K-means clustering with either $k = 2$ or $k = 3$. This selection process prioritized features most relevant for [UFC](#), consequently omitting features important for [PBC](#). The number of selected features is presented in Table 2, the fourth column ($|\mathcal{F}^*|$), for both methods. The classification is repeated with the same classifiers using the same testing conditions with the only difference being the number of features. Additionally, we computed $\Delta\mathcal{A}$ as the difference between the accuracy $\mathcal{A}$ computed with all initial features $\mathcal{F}$ and the accuracy $\mathcal{A}^*$ with the selected features $\mathcal{F}^*$. As observed in Table 2, the classification accuracy $\mathcal{A}^*$ is lower than $\mathcal{A}$ for most classification criteria.

As our research goal (Eq. 2), we aimed to discover a feature set $\mathcal{F}^*$ with significantly reduced classification accuracy for [PBC](#) compared to the original feature set, while ensuring that the accuracy of [UFC](#) remains at desired levels. These targeted accuracy levels are to be determined by the data owners.

The proposed approach demonstrates excellent results on the UAH dataset by significantly reducing the number of features to be publicly released from 17 to 2 (for [IDS](#)-based clustering) and to 1 (for K-means clustering). Additionally, it led to a decrease in the accuracy of driver classification by approximately 0.50 for all tested classifiers (FCN-LSTM, SVM, and XGBoost) with [IDS](#)-based clustering, and by over 0.66 for K-means clustering. Despite these reductions, the accuracy of road-type identification remained consistently above 0.99. Therefore, it is sufficient to release only one feature to preserve user privacy while maintaining the utility of road-type detection. Comparable results were obtained for the HCRL dataset concerning driver/road-type classification. Regarding the behavior detection utility objective, the approach successfully decreased the number of features from 17 to 7 (with [IDS](#)-based clustering) and 13 (with

Table 2: Classification results for the selected set of features using IDS-based clustering and K-means clustering.

Clustering method	Dataset	$ \mathcal{F} $	Utility classification objective(s)	$ \mathcal{F}^* $	Classification	FCN-LSTM		SVM		XGBoost	
						$\mathcal{A}^*$	$\Delta\mathcal{A}$	$\mathcal{A}^*$	$\Delta\mathcal{A}$	$\mathcal{A}^*$	$\Delta\mathcal{A}$
IDS ($\theta_I = 0.0$)	UAH	17	Road	2	Driver (PBC)	0.4133	-0.5023	0.2388	-0.5047	0.4822	-0.4948
					Road (UFC)	0.9993	-0.0006	0.9974	-0.0019	0.9997	-0.0002
			Behavior	7	Driver (PBC)	0.6958	-0.2198	0.4632	-0.2803	0.7103	-0.2667
					Behavior (UFC)	0.8420	-0.1042	0.7280	-0.1177	0.8409	-0.1376
			Road and Behavior	9	Driver (PBC)	0.8756	-0.0400	0.5315	-0.2120	0.8382	-0.1388
					Behavior (UFC)	0.9171	-0.0291	0.7599	-0.0858	0.9023	-0.0762
	HCRL	15	Road	5	Road (UFC)	0.9996	-0.0003	0.9980	-0.0013	0.9998	-0.0001
					Driver (PBC)	0.2428	-0.6702	0.2029	-0.5704	0.2581	-0.6707
	MotionSense	12	Activity	6	Road (UFC)	0.9140	-0.0496	0.9038	-0.0243	0.9165	-0.0516
					User (PBC)	0.5987	-0.2605	0.4345	-0.2721	0.5588	-0.2194
	PAMAP2	31	Activity	24	Activity (UFC)	0.7597	-0.1557	0.7056	-0.1303	0.7346	-0.1020
					User (PBC)	0.9715	-0.0265	0.5187	-0.4810	0.9523	-0.0477
K-means ($k = 2$)	UAH	17	Road	1	Activity (UFC)	0.9651	-0.0002	0.5986	-0.3692	0.9513	-0.0460
					Driver (PBC)	0.2490	-0.6666	0.2321	-0.5114	0.2636	-0.7134
			Behavior	13	Road (UFC)	0.9903	-0.0096	0.9900	-0.0093	0.9899	-0.0100
					Driver (PBC)	0.6009	-0.3147	0.4594	-0.2841	0.5794	-0.3976
			Road and Behavior	14	Behavior (UFC)	0.7567	-0.1895	0.6640	-0.1817	0.7457	-0.2328
					Driver (PBC)	0.6566	-0.2590	0.5051	-0.2384	0.6990	-0.2780
	HCRL	15	Road	3	Behavior (UFC)	0.7809	-0.1653	0.6809	-0.1648	0.8038	-0.1747
					Road (UFC)	0.9978	-0.0021	0.9948	-0.0045	0.9994	-0.0005
	MotionSense	12	Activity	6	Driver (PBC)	0.1823	-0.7307	0.1687	-0.6046	0.1671	-0.7617
					Road (UFC)	0.8887	-0.0749	0.8850	-0.0431	0.8881	-0.0800
	PAMAP2	31	Activity	28	User (PBC)	0.5987	-0.2605	0.4345	-0.2721	0.5588	-0.2194
					Activity (UFC)	0.7597	-0.1557	0.7056	-0.1303	0.7346	-0.1020
K-means ($k = 3$)	PAMAP2	31	Activity	3	User (PBC)	0.9701	-0.0271	0.9452	-0.0545	0.9523	-0.0477
					Activity (UFC)	0.9825	0.0172	0.9644	-0.0353	0.9513	-0.0460

K-means clustering), while the accuracy of driver detection decreased by more than 0.21 across all tested scenarios. Notably, behavior detection was found to be higher for features selected by IDS-based clustering compared to those selected by K-means clustering.

For the MotionSense dataset, the features were reduced from 12 to 6 using both clustering methods, with the same set of selected features. This reduction led to a decrease in user identification classification accuracy to below 0.60. Concurrently, the utility accuracy also decreased to approximately 0.75.

The PAMAP2 dataset posed a challenge due to the remarkably high classification rate for the PBC (0.99), which is slightly higher than that of the UFC (0.96). By applying the IDS-based clustering method, there was a minimal reduction in the number of features. Similarly, the K-means method with $k = 2$ did not produce significant reductions. However, for $k = 3$, the number of selected features dramatically decreased to three, with the UFC achieving higher accuracy than the PBC. Nonetheless, both accuracies remained relatively low (0.65–0.70). Consequently, the proposed method could offer limited applicability for the PAMAP2 dataset, suggesting the need for alternative approaches for feature selection or introducing DP-based perturbation.

Furthermore, the case of multiple UFCs was tested for the UAH dataset with two utility classification objectives: behavior and road type detection simultaneously. In this case, the features were selected as described in Section 4.3. Significantly better results were obtained for the K-means clustering method, illustrating that in the case of multiple UFCs, the method of feature selection requires further research.

6. Conclusions

The current research proposed a novel privacy-oriented feature selection approach targeted for multivariate time series data in the context of **TSC**. By defining the problem in the context of two opposing classifiers (**UFC** and **PBC**) and by introducing the **IDS**, the methodology identifies and selects features that are important for user-specific classification while minimizing their applicability to potential adversaries. Moreover, the proposed approach is configurable, allowing the data owners to achieve a balance between privacy preservation and data utility. The technique is particularly valuable in scenarios where altering data to protect privacy is not possible due to the necessity of preserving data precision. The proposed methodology was evaluated in diverse scenarios, including driver and **HAR** datasets. Experimental results demonstrated that our approach effectively balances privacy and utility for certain datasets within the context of **TSC**. Nonetheless, for other datasets, the proposed feature selection proved not to be sufficient, and other protection methods should be implemented. However, even for these datasets, we consider the approach a useful data exploratory step for better understanding the data and, equally important, the capabilities of potential attackers.

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